Deep Reinforcement Learning for Conversational AI

Hands-on Tutorial

Om Sonie Abir Chakraborty Ankan Mullick

Agenda:

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Deep Models [1-3]

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Deep Reinforcement Learning for Dialogue Generation [4]
Dialog control with Supervised and Reinforcement Learning [5]
A Deep Reinforcement Learning Chatbot [6]
Deep Reinforcement Learning for Task-Oriented Dialogue [7]
Deep Reinforcement Learning for Sequence-to-Sequence Models [8]
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Natural Language Generation Using RL [11]
Hierarchical RL for Open-Domain Dialog [12]
Goal-oriented Chatbots with Hierarchical RL [13]
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Deep Dyna-Q: Integrating Planning for Task-Completion Dialogue Policy Learning [15]

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CONCLUSION (5min)

0.
Deep Reinforcement Learning Foundation (45min)

Deep Reinforcement Learning Foundation:

- Reinforcement Learning [1-3]
- Tabular Value Function [1-3]
- Deep Value Functions [1-3]
- Deep Policies [1-3]
- Deep Models [1-3]

1.
Deep Reinforcement Learning Approaches-I (45min)

Deep Reinforcement Learning Approaches-I:

- Deep Reinforcement Learning for Dialogue Generation [4]
- Dialog control with Supervised and Reinforcement Learning [5]
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- Deep Reinforcement Learning for Conversational AI [9]

1.1 Deep Reinforcement Learning for Dialogue Generation

Recent neural models of dialogue generation offer great promise for generating responses for conversational agents, but tend to be shortsighted, predicting utterances one at a time while ignoring their influence on future outcomes.

Modeling the future direction of a dialogue is crucial to generating coherent, interesting dialogues, a need which led traditional NLP models of dialogue to draw on reinforcement learning. In this paper, we show how to integrate these goals, applying deep reinforcement learning to model future reward in chatbot dialogue. The model simulates dialogues between two virtual agents, using policy gradient methods to reward sequences that display three useful conversational properties: informativity, coherence, and ease of answering (related to forward-looking function).

We evaluate our model on diversity, length as well as with human judges, showing that the proposed algorithm generates more interactive responses and manages to foster a more sustained conversation in dialogue simulation.

This work marks a first step towards learning a neural conversational model based on the long-term success of dialogues.

[4] Jiwei Li., et. al. Deep Reinforcement Learning for Dialogue Generation

1.1 Deep Reinforcement Learning for Dialogue Generation

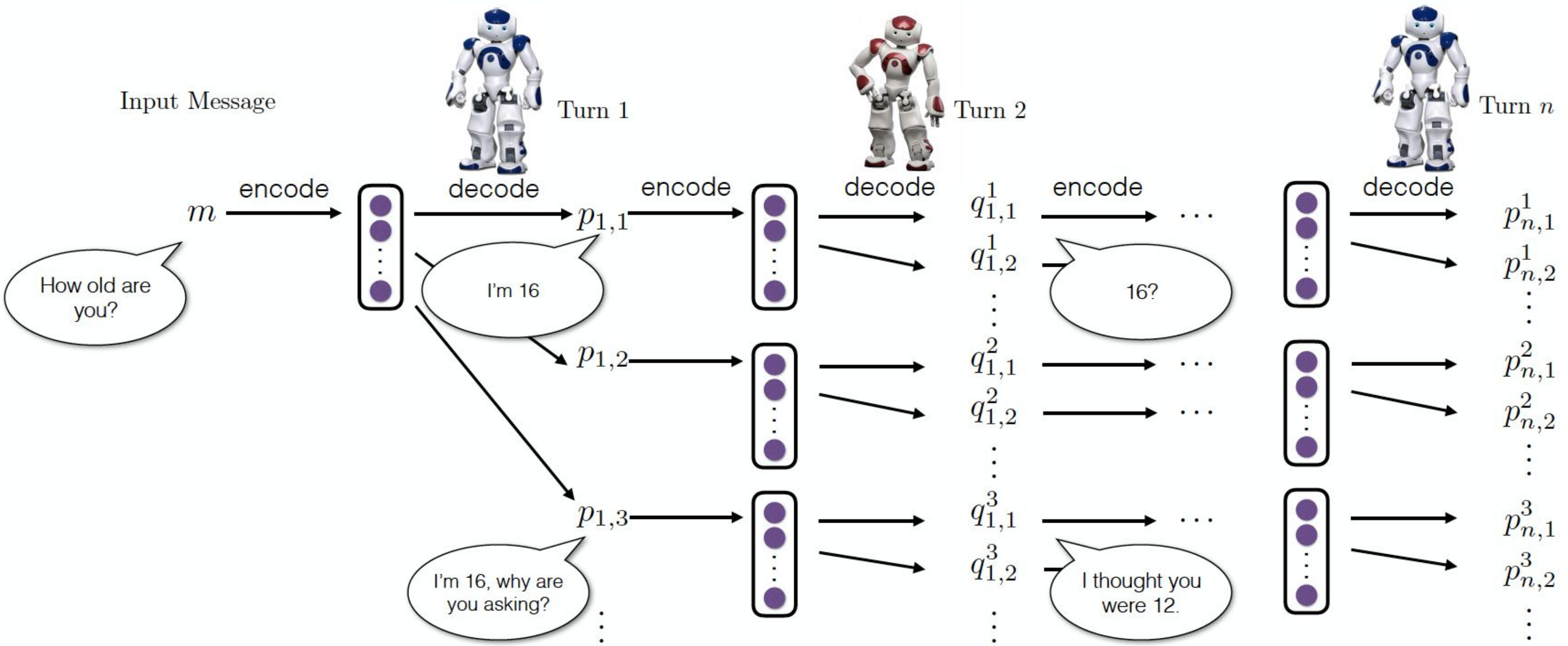


Figure 1: Dialogue simulation between the two agents.

[4] Jiwei Li., et. al. Deep Reinforcement Learning for Dialogue Generation

1.2 Hybrid Code Networks: practical and efficient end-to-end dialog control with supervised and reinforcement learning

End-to-end learning of recurrent neural networks (RNNs) is an attractive solution for dialog systems; however, current techniques are data-intensive and require thousands of dialogs to learn simple behaviors. We introduce Hybrid Code Networks (HCNs), which combine an RNN with domain-specific knowledge encoded as software and system action templates.

Compared to existing end-toend approaches, HCNs considerably reduce the amount of training data required, while retaining the key benefit of inferring a latent representation of dialog state. In addition, HCNs can be optimized with supervised learning, reinforcement learning, or a mixture of both. HCNs attain stateof-the-art performance on the bAbI dialog dataset (Bordes and Weston, 2016), and outperform two commercially deployed customer-facing dialog systems.

[5] Jason D. et. al., Hybrid Code Networks: practical and efficient end-to-end dialog control with supervised and reinforcement learning

1.2 Hybrid Code Networks: practical and efficient end-to-end dialog control with supervised and reinforcement learning

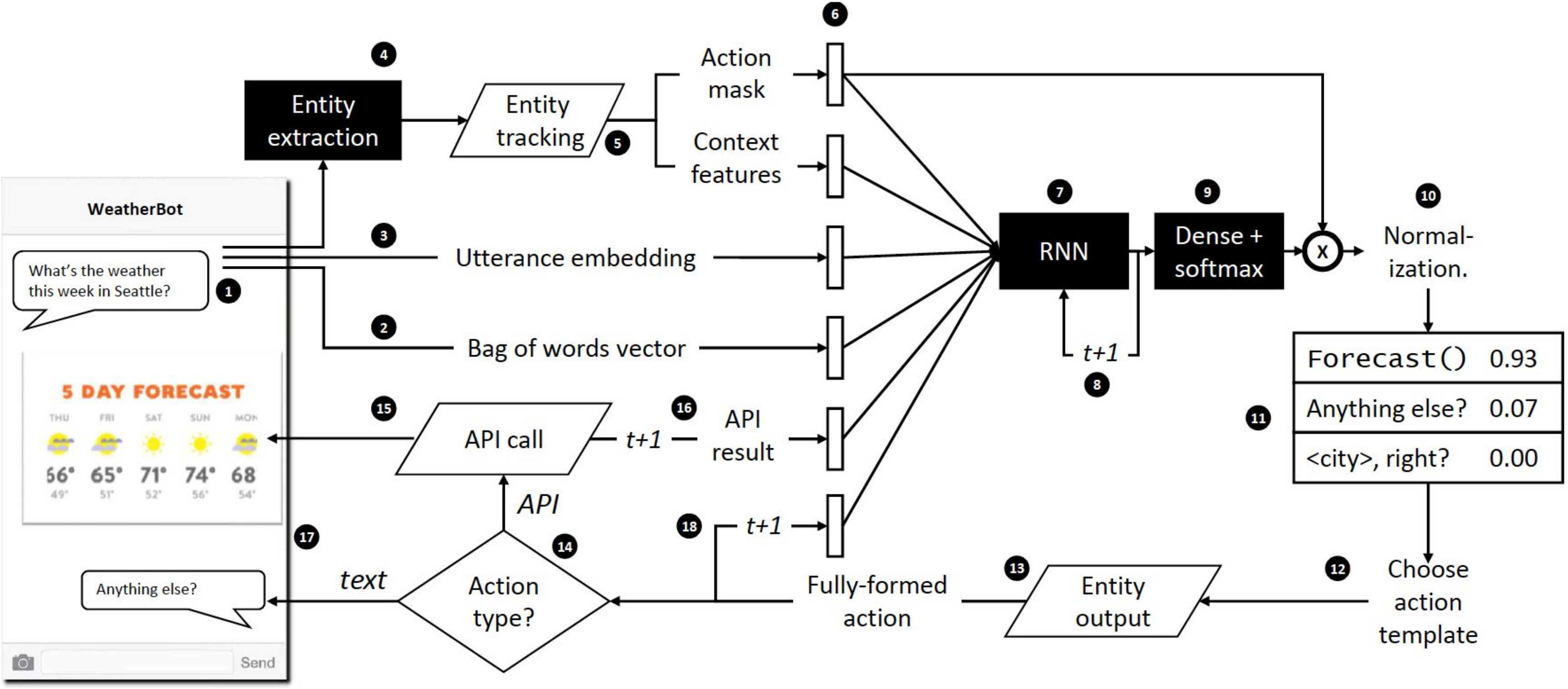


Figure 1: Operational loop. Trapezoids refer to programmatic code provided by the software developer, and shaded boxes are trainable components. Vertical bars under "6" represent concatenated vectors which form the input to the RNN.

[5] Jason D. et. al., Hybrid Code Networks: practical and efficient end-to-end dialog control with supervised and reinforcement learning

1.3 A Deep Reinforcement Learning Chatbot

We present MILABOT: a deep reinforcement learning chatbot developed by the Montreal Institute for Learning Algorithms (MILA) for the Amazon Alexa Prize competition. MILABOT is capable of conversing with humans on popular small talk topics through both speech and text.

The system consists of an ensemble of natural language generation and retrieval models, including template-based models, bag-of-words models, sequence-to-sequence neural network and latent variable neural network models. By applying reinforcement learning to crowdsourced data and real-world user interactions, the system has been trained to select an appropriate response from the models in its ensemble.

The system has been evaluated through A/B testing with real-world users, where it performed significantly better than many competing systems. Due to its machine learning architecture, the system is likely to improve with additional data.

[6] Iulian V. Serban, et. al., A Deep Reinforcement Learning Chatbot

1.3 A Deep Reinforcement Learning Chatbot

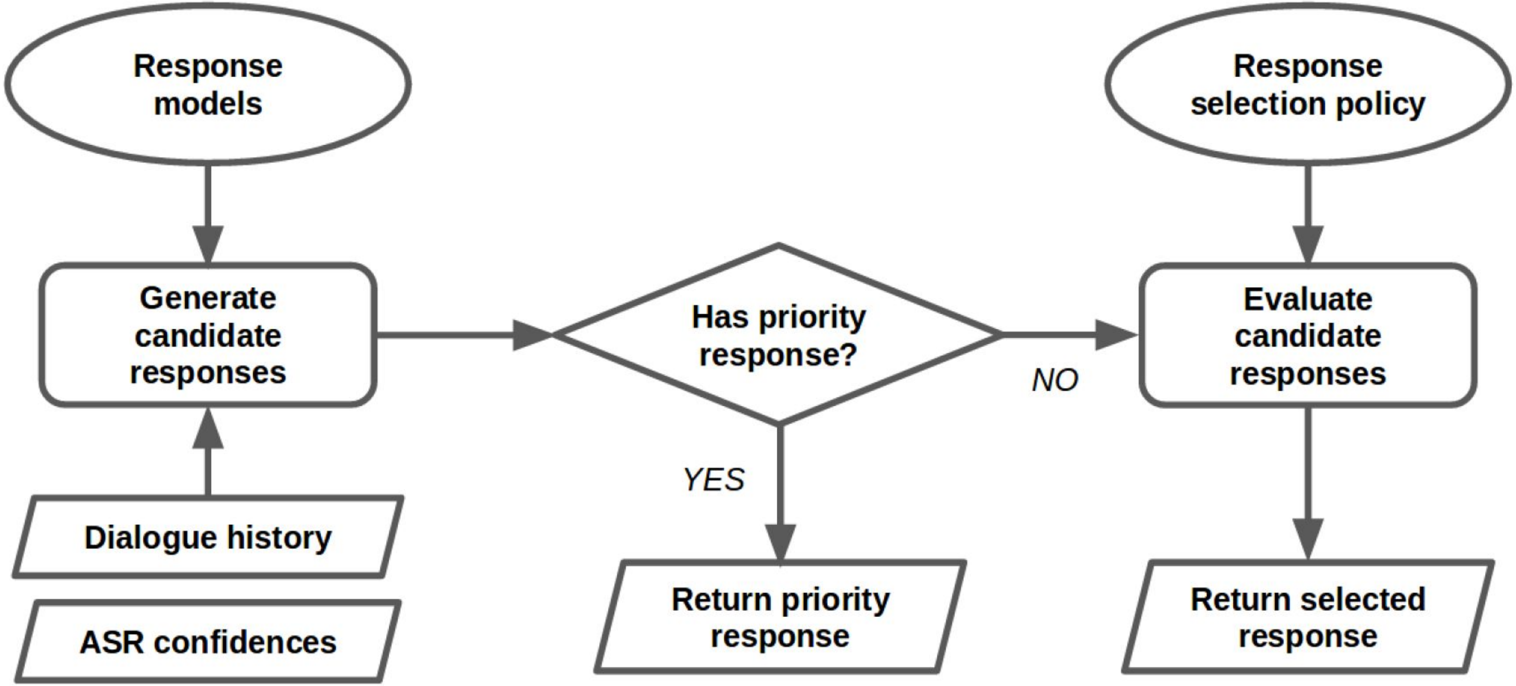


Figure 1: Dialogue manager control flow

[6] Iulian V. Serban, et. al., A Deep Reinforcement Learning Chatbot

1.3 A Deep Reinforcement Learning Chatbot

Add Architecture details

[6] Iulian V. Serban, et. al., A Deep Reinforcement Learning Chatbot

1.4 Deep Reinforcement Learning for Task-Oriented Dialogue

Despite major advances in NLP over the past decade, most deployed task-oriented chatbots are implemented as a hand-coded Finite State Machine (FSM) architecture, where user utterances matching pre-defined patterns transition the dialogue from one state to the next.

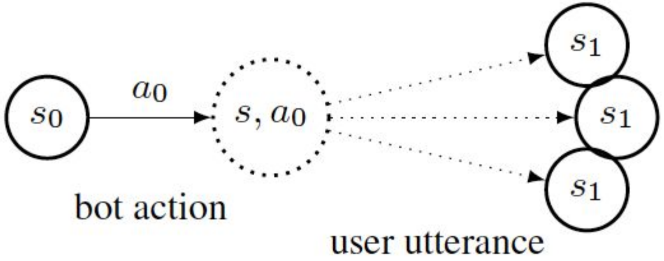
The open-source framework Rasa (rasa.com) has popularized a new architecture for chatbots with flexible dialogue states and transitions that can be learned [1] rather than pre-defined. However, there exists a training data problem within this new architecture.

Rasa requires training data for both NLU (intent recognition and parsing) and dialogue state management, but hand-annotating examples for training is a time-consuming and expensive process. Moreover, there is no simple way to leverage the conversation data generated by chatbots deployed in the field. In this paper, we demonstrate the viability of end-to-end Deep Reinforcement Learning for automatic and continuous learning for a chatbot agent.

We describe the process of building a machine-learned action policy model for a chat agent to approximate the behavior of a deployed system, heuristics and conversational cues to extract rewards from unannotated conversation logs, and finally the use of Reinforcement Learning to improve performance over our baseline.

[7] Nnamdi Iregbulem, et. al., Deep Reinforcement Learning for Task-Oriented Dialogue

1.4 Deep Reinforcement Learning for Task-Oriented Dialogue



[7] Nnamdi Iregbulem, et. al., Deep Reinforcement Learning for Task-Oriented Dialogue

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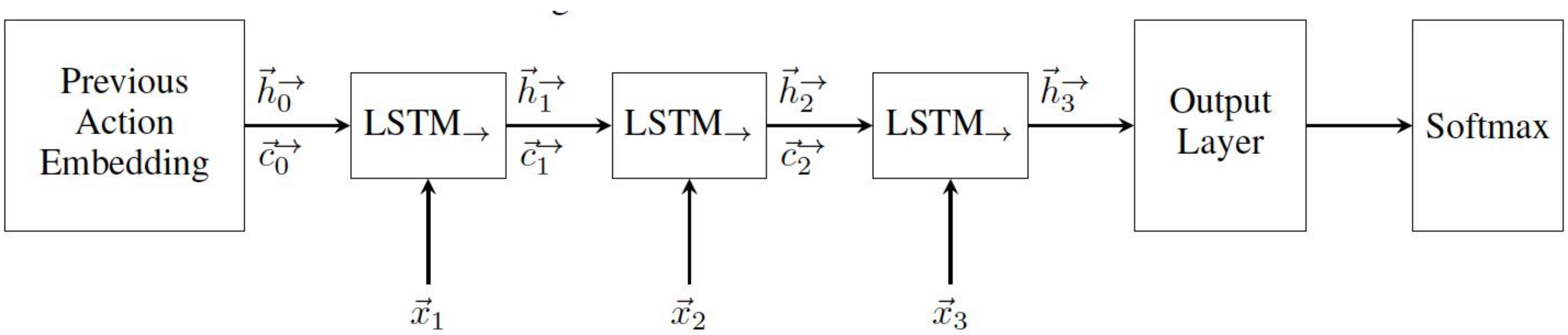


Figure 1: Model Architecture

[7] Nnamdi Iregbulem, et. al., Deep Reinforcement Learning for Task-Oriented Dialogue

1.5 Deep Reinforcement Learning for Sequence-to-Sequence Models

In recent times, sequence-to-sequence (seq2seq) models have gained a lot of popularity and provide state-ofthe-art performance in a wide variety of tasks such as machine translation, headline generation, text summarization, speech to text conversion, and image caption generation. The underlying framework for all these models is usually a deep neural network comprising an encoder and a decoder.

Although simple encoderdecoder models produce competitive results, many researchers have proposed additional improvements over these seq2seq models, e.g., using an attention-based model over the input, pointergeneration models, and self-attention models. However, such seq2seq models suffer from two common problems: 1) exposure bias and 2) inconsistency between train/test measurement.

Recently, a completely novel point of view has emerged in addressing these two problems in seq2seq models, leveraging methods from reinforcement learning (RL). In this survey, we consider seq2seq problems from the RL point of view and provide a formulation combining the power of RL methods in decision-making with seq2seq models that enable remembering long-term memories.

We present some of the most recent frameworks that combine concepts from RL and deep neural networks. Our work aims to provide insights into some of the problems that inherently arise with current approaches and how we can address them with better RL models.

We also provide the source code for implementing most of the RL models discussed in this paper to support the complex task of abstractive text summarization and provide some targeted experiments for these RL models, both in terms of performance and training time.

[8] Yaser K., et. al., Deep Reinforcement Learning for Sequence-to-Sequence Models

1.5 Deep Reinforcement Learning for Sequence-to-Sequence Models

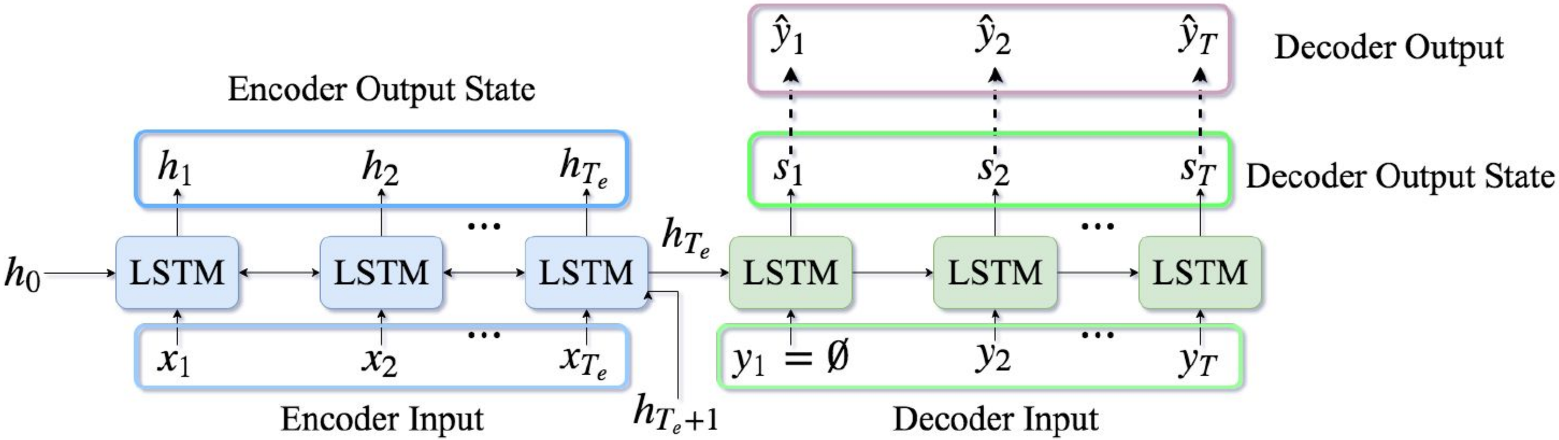


Fig. 1: A simple seq2seq model. The blue boxes correspond to the encoder part which has Te units. The green boxes correspond to the decoder part which has T units.

[8] Yaser K., et. al., Deep Reinforcement Learning for Sequence-to-Sequence Models

1.6 Deep Reinforcement Learning for Conversational AI

Deep reinforcement learning is revolutionizing the artificial intelligence field. Currently, it serves as a good starting point for constructing intelligent autonomous systems which offer a better knowledge of the visual world.

It is possible to scale deep reinforcement learning with the use of deep learning and do amazing tasks such as use of pixels in playing video games. In this paper, key concepts of deep reinforcement learning including reward function, differences between reinforcement learning and supervised learning and models for implementation of reinforcement are discussed.

Key challenges related to the implementation of reinforcement learning in conversational AI domain are identified as well as discussed in detail.

Various conversational models which are based on deep reinforcement learning (as well as deep learning) are also discussed. In summary, this paper discusses key aspects of deep reinforcement learning which are crucial for designing an efficient conversational AI.

[9] Mahipal J., et. al., Deep Reinforcement Learning for Conversational AI

1.6 Deep Reinforcement Learning for Conversational AI

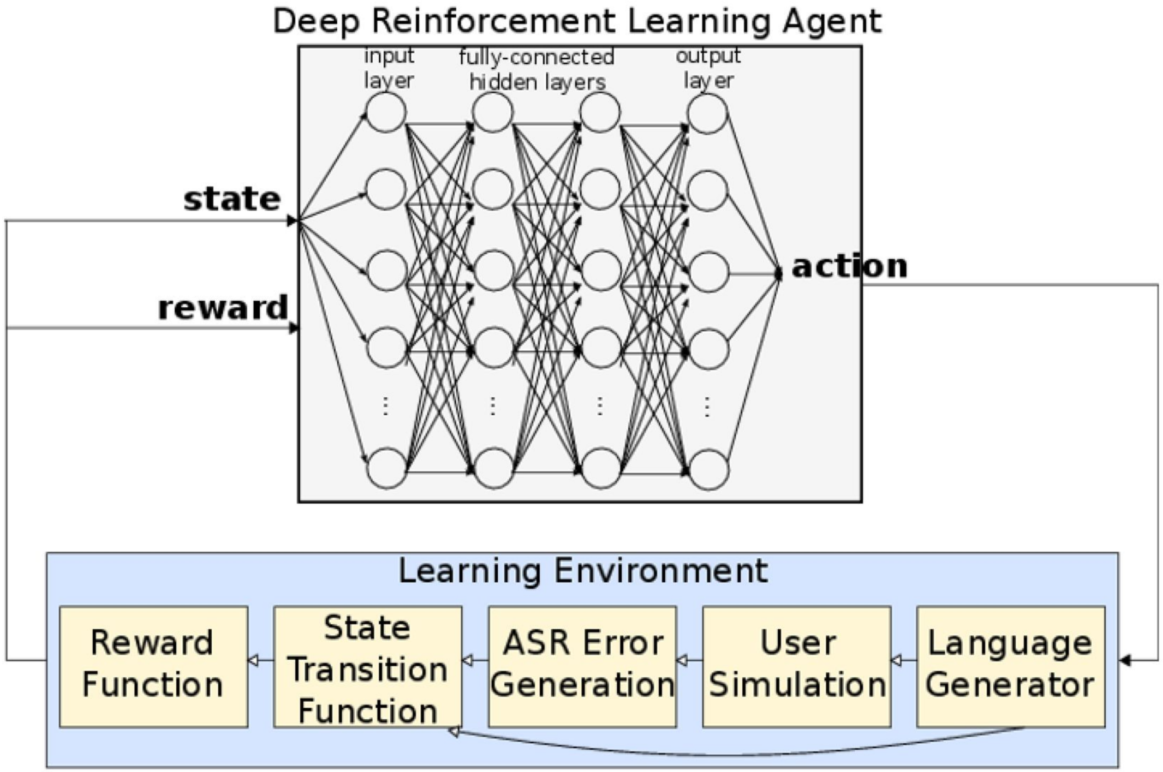


Figure 1: Deep Reinforcement Learning Agent 1

[9] Mahipal J., et. al., Deep Reinforcement Learning for Conversational AI

1.6 Deep Reinforcement Learning for Conversational AI

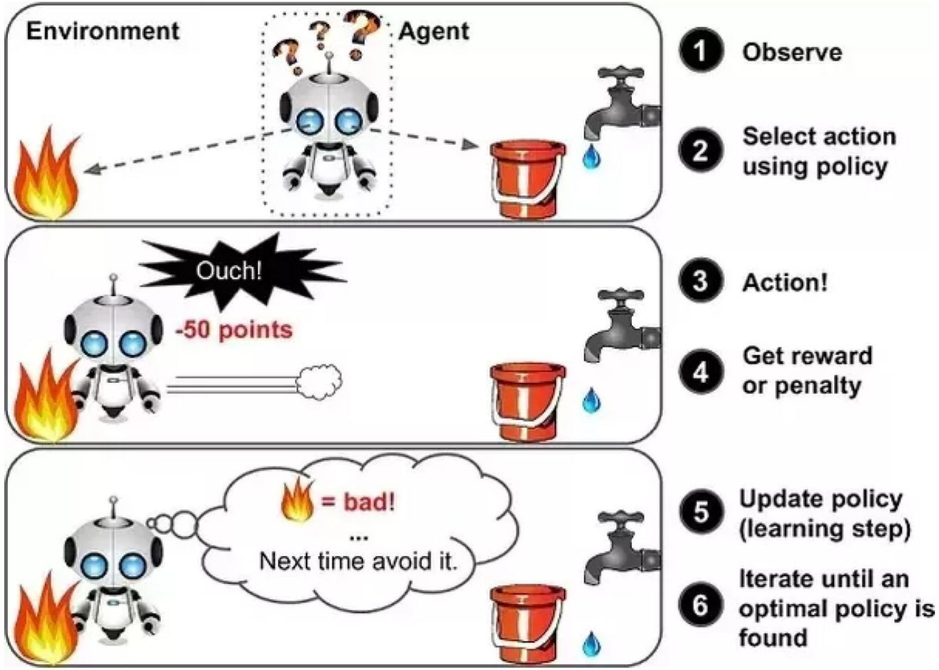


Figure 2: Summary of Reinforcement Learning

[9] Mahipal J., et. al., Deep Reinforcement Learning for Conversational AI

2.
Deep Reinforcement Learning Approaches-II (45min)

Deep Reinforcement Learning Approaches-II:

- Deep Reinforcement Learning For Modeling Chit-Chat Dialog [10]
- Natural Language Generation Using RL [11]
- Hierarchical RL for Open-Domain Dialog [12]
- Goal-oriented Chatbots with Hierarchical RL [13]
- Human-centric dialog training via offline RL [14]
- Deep Dyna-Q: Integrating Planning for Task-Completion Dialogue Policy Learning [15]

2.1 Deep RL For Modeling Chit-Chat Dialog With Discrete Attributes

Open domain dialog systems face the challenge of being repetitive and producing generic responses. In this paper, we demonstrate that by conditioning the response generation on interpretable discrete dialog attributes and composed attributes, it helps improve the model perplexity and results in diverse and interesting non-redundant responses.

We propose to formulate the dialog attribute prediction as a reinforcement learning (RL) problem and use policy gradients methods to optimize utterance generation using long-term rewards.

Unlike existing RL approaches which formulate the token prediction as a policy, our method reduces the complexity of the policy optimization by limiting the action space to dialog attributes, thereby making the policy optimization more practical and sample efficient.

We demonstrate this with experimental and human evaluations.

[10] Chinnadhurai S. , et. al. Deep Reinforcement Learning For Modeling Chit-Chat Dialog With Discrete Attributes

2.1 Deep RL For Modeling Chit-Chat Dialog With Discrete Attributes

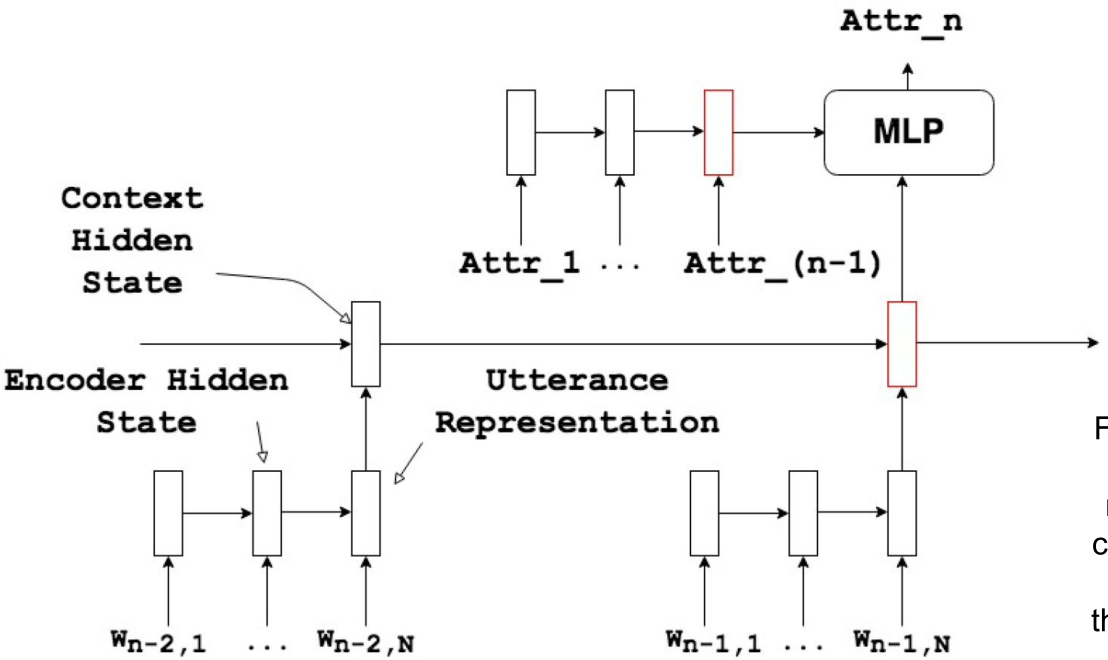


Figure 1: Dialog attribute classification: We predict the dialog attribute of the next utterance based on the previous context and attributes corresponding to the previous utterances. Please note that we depict only a single attribute for convenience

[10] Chinnadhurai S. , et. al. Deep Reinforcement Learning For Modeling Chit-Chat Dialog With Discrete Attributes

2.1 Deep RL For Modeling Chit-Chat Dialog With Discrete Attributes

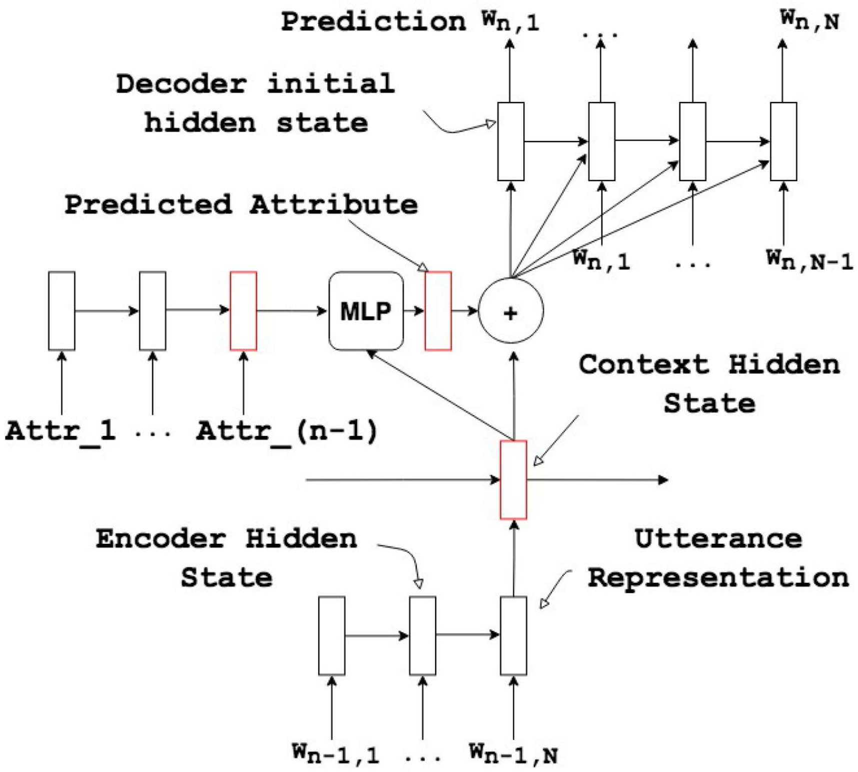


Figure 2: Attribute Conditional HRED : Token generation is additionally conditioned on the predicted dialog attributes. The dialog attribute's embedding is concatenated with the context vector.

[10] Chinnadhurai S. , et. al. Deep Reinforcement Learning For Modeling Chit-Chat Dialog With Discrete Attributes

2.2 *Natural Language Generation Using RL with External Rewards*

We propose an approach towards natural language generation using bidirectional encoder-decoder which incorporates external rewards through reinforcement learning (RL).

We use attention mechanism and maximum mutual information as initial objective function using RL. Using a two-part training scheme, we train an external reward analyzer to predict the external rewards and then use the predicted rewards to maximize the expected rewards (both internal and external).

We evaluate the system on two standard dialogue corpora - Cornell Movie Dialog Corpus and Yelp Restaurant Review Corpus.

We report standard evaluation metrics including BLEU, ROUGE-L and perplexity as well as human evaluation to validate our approach.

[11] Vidhushini S., et. al., Natural Language Generation Using Reinforcement Learning with External Rewards

2.2 *Natural Language Generation Using RL with External Rewards*

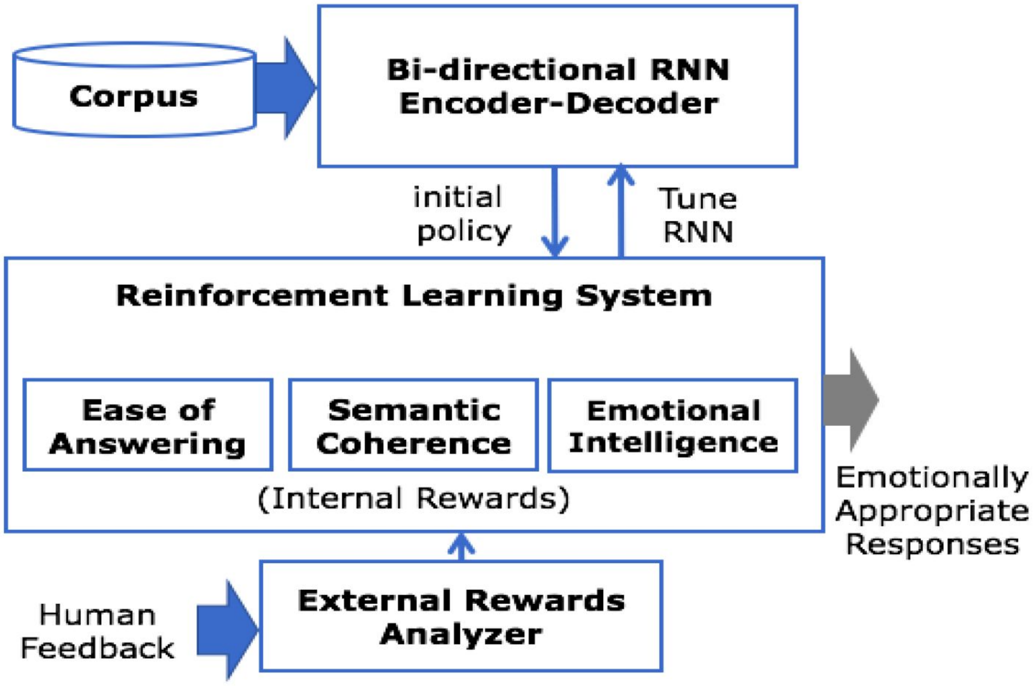


Fig. 1: Overall Architecture of the system showing internal and external rewards using reinforcement learning

[11] Vidhushini S., et. al., Natural Language Generation Using Reinforcement Learning with External Rewards

2.3 Hierarchical Reinforcement Learning for Open-Domain Dialog

Open-domain dialog generation is a challenging problem; maximum likelihood training can lead to repetitive outputs, models have difficulty tracking long-term conversational goals, and training on standard movie or online datasets may lead to the generation of inappropriate, biased, or offensive text.

Reinforcement Learning (RL) is a powerful framework that could potentially address these issues, for example by allowing a dialog model to optimize for reducing toxicity and repetitiveness. However, previous approaches which apply RL to open-domain dialog generation do so at the word level, making it difficult for the model to learn proper credit assignment for long-term conversational rewards.

In this paper, we propose a novel approach to hierarchical reinforcement learning (HRL), VHRL, which uses policy gradients to tune the utterance-level embedding of a variational sequence model. This hierarchical approach provides greater flexibility for learning long-term, conversational rewards.

We use self play and RL to optimize for a set of human-centered conversation metrics, and show that our approach provides significant improvements – in terms of both human evaluation and automatic metrics – over state-of-the-art dialog models, including Transformers.

[12] Abdelrhman S., et. al., Hierarchical Reinforcement Learning for Open-Domain Dialog

2.3 Hierarchical Reinforcement Learning for Open-Domain Dialog

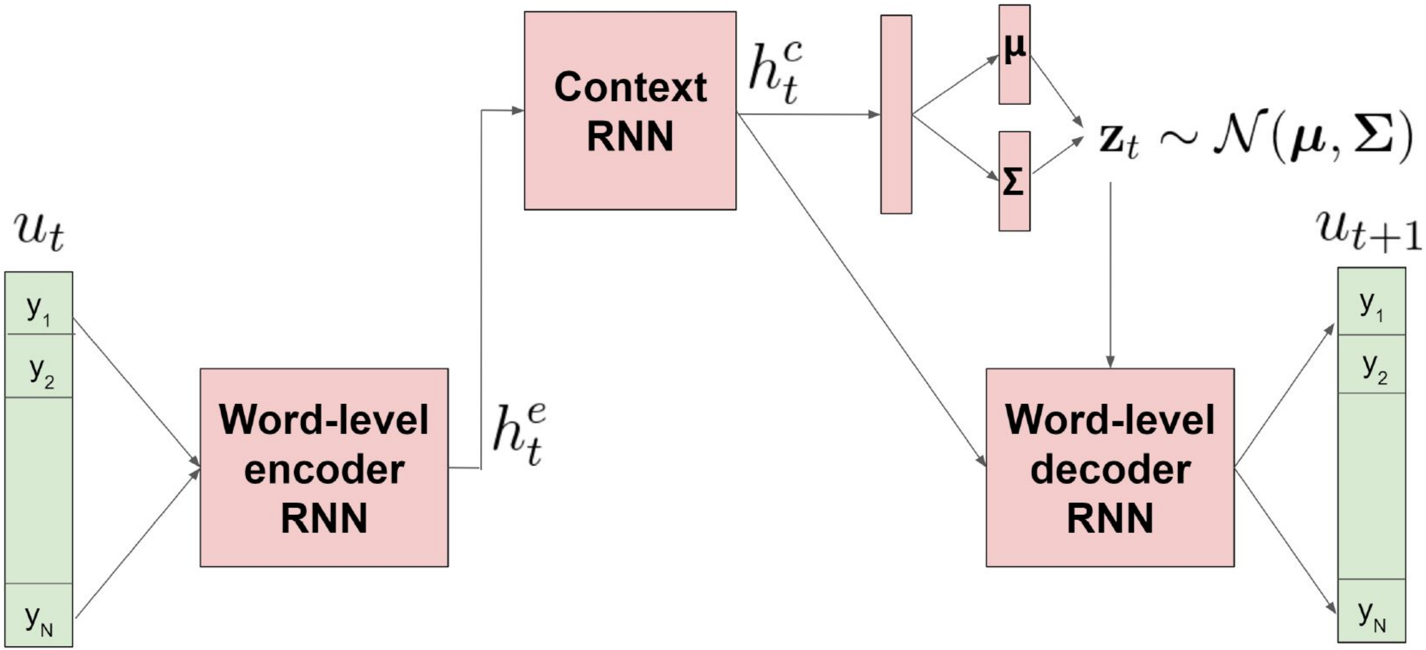


Figure 1: VHRED model architecture, where the embedding vector z for each utterance is sampled from a multivariate normal distribution using the reparameterization trick.

[12] Abdelrhman S., et. al., Hierarchical Reinforcement Learning for Open-Domain Dialog

2.4 GoChat: Goal-oriented Chatbots with Hierarchical Reinforcement Learning

A chatbot that converses like a human should be goal-oriented (i.e., be purposeful in conversation), which is beyond language generation. However, existing dialogue systems often heavily rely on cumbersome hand-crafted rules or costly labelled datasets to reach the goals.

In this paper, we propose Goal-oriented Chatbots (GoChat), a framework for end-to-end training chatbots to maximize the long term return from offline multi-turn dialogue datasets.

Our framework utilizes hierarchical reinforcement learning (HRL), where the high-level policy guides the conversation towards the final goal by determining some sub-goals, and the low-level policy fulfills the sub-goals by generating the corresponding utterance for response.

In our experiments on a real-world dialogue dataset for anti-fraud in financial, our approach outperforms previous methods on both the quality of response generation as well as the success rate of accomplishing the goal.

[13] Jianfeng L et. al., GoChat: Goal-oriented Chatbots with Hierarchical Reinforcement Learning

2.4 GoChat: Goal-oriented Chatbots with Hierarchical Reinforcement Learning

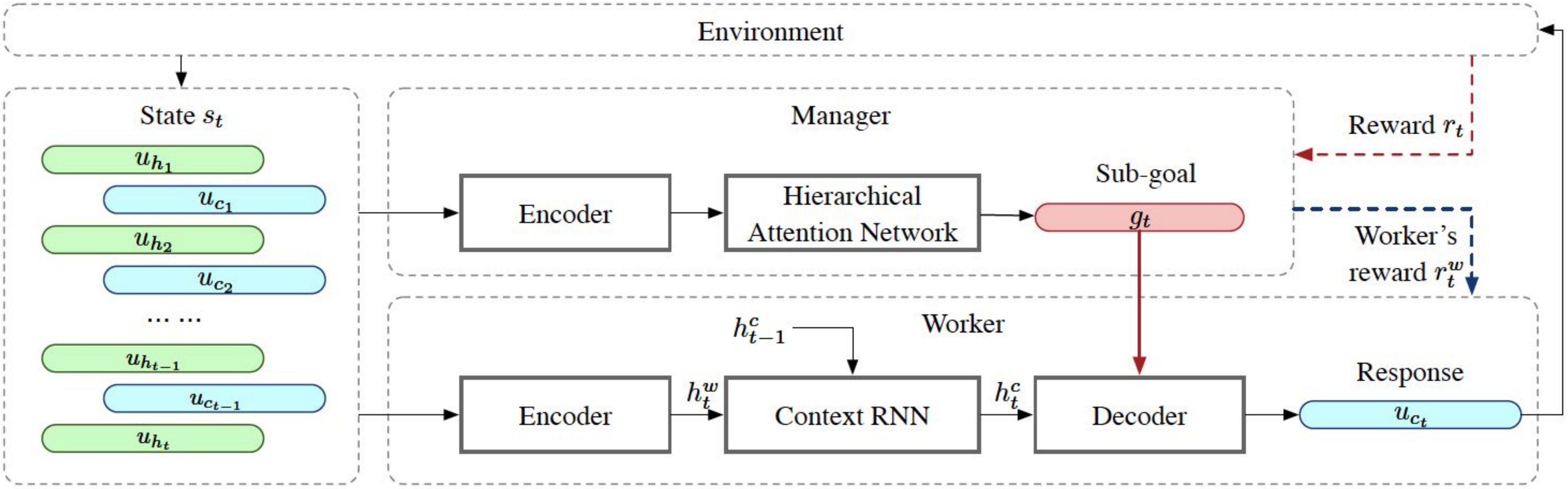


Figure 1: The overall GoChat framework for goal-oriented chatbots with hierarchical reinforcement learning. The manager decides the sub-goals for the worker, and the worker generates the response according to the observations and the sub-goal.

[13] *Jianfeng L et. al., GoChat: Goal-oriented Chatbots with Hierarchical Reinforcement Learning*

2.5 Human-centric dialog training via offline reinforcement learning

How can we train a dialog model to produce better conversations by learning from human feedback, without the risk of humans teaching it harmful chat behaviors? We start by hosting models online, and gather human feedback from real-time, open-ended conversations, which we then use to train and improve the models using offline reinforcement learning (RL).

We identify implicit conversational cues including language similarity, elicitation of laughter, sentiment, and more, which indicate positive human feedback, and embed these in multiple reward functions. A well known challenge is that learning an RL policy in an offline setting usually fails due to the lack of ability to explore and the tendency to make over-optimistic estimates of future reward.

These problems become even harder when using RL for language models, which can easily have a 20,000 action vocabulary and many possible reward functions. We solve the challenge by developing a novel class of offline RL algorithms. These algorithms use KL-control to penalize divergence from a pre-trained prior language model, and use a new strategy to make the algorithm pessimistic, instead of optimistic, in the face of uncertainty.

We test the resulting dialog model with ratings from 80 users in an open-domain setting and find it achieves significant improvements over existing deep offline RL approaches. The novel offline RL method is viable for improving any existing generative dialog model using a static dataset of human feedback.

[14] Natasha J., et. al., Human-centric dialog training via offline reinforcement learning

2.5 Human-centric dialog training via offline reinforcement learning

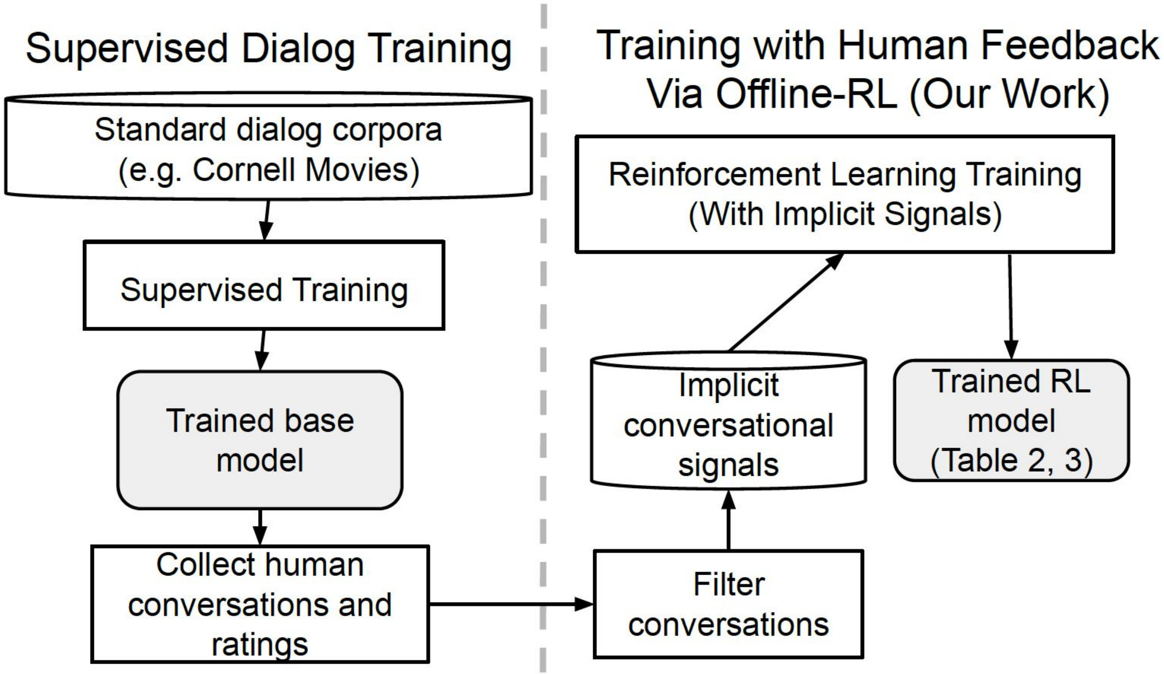


Figure 1: Schematic diagram of our method for training with human conversation cues via offline RL. Unlike traditional approaches which stop at using explicit feedback to evaluate static conversations, we allow humans to freely interact with dialog models, and compute metrics based on their implicit satisfaction which are optimized using offline RL.

[14] Natasha J., et. al., Human-centric dialog training via offline reinforcement learning

2.6 Deep Dyna-Q: Integrating Planning for Task-Completion Dialogue Policy Learning

Training a task-completion dialogue agent via reinforcement learning (RL) is costly because it requires many interactions with real users.

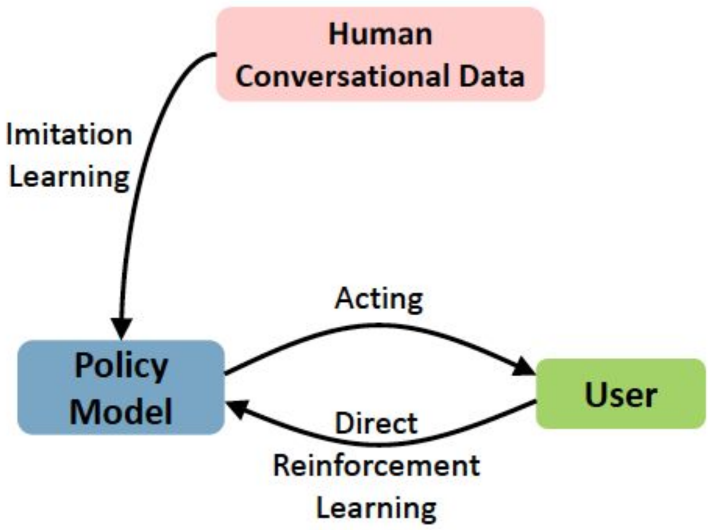
One common alternative is to use a user simulator. However, a user simulator usually lacks the language complexity of human interlocutors and the biases in its design may tend to degrade the agent. To address these issues, we present Deep Dyna-Q, which to our knowledge is the first deep RL framework that integrates planning for task-completion dialogue policy learning.

We incorporate into the dialogue agent a model of the environment, referred to as the world model, to mimic real user response and generate simulated experience. During dialogue policy learning, the world model is constantly updated with real user experience to approach real user behavior, and in turn, the dialogue agent is optimized using both real experience and simulated experience.

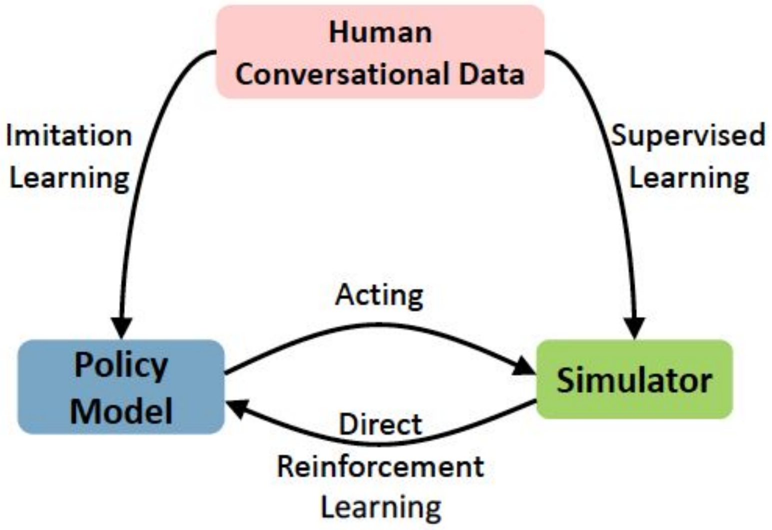
The effectiveness of our approach
is demonstrated on a movie-ticket booking task in both simulated and human-in-the loop settings.

[15] Baolin P, et. al., Deep Dyna-Q: Integrating Planning for Task-Completion Dialogue Policy Learning

2.6 Deep Dyna-Q: Integrating Planning for Task-Completion Dialogue Policy Learning



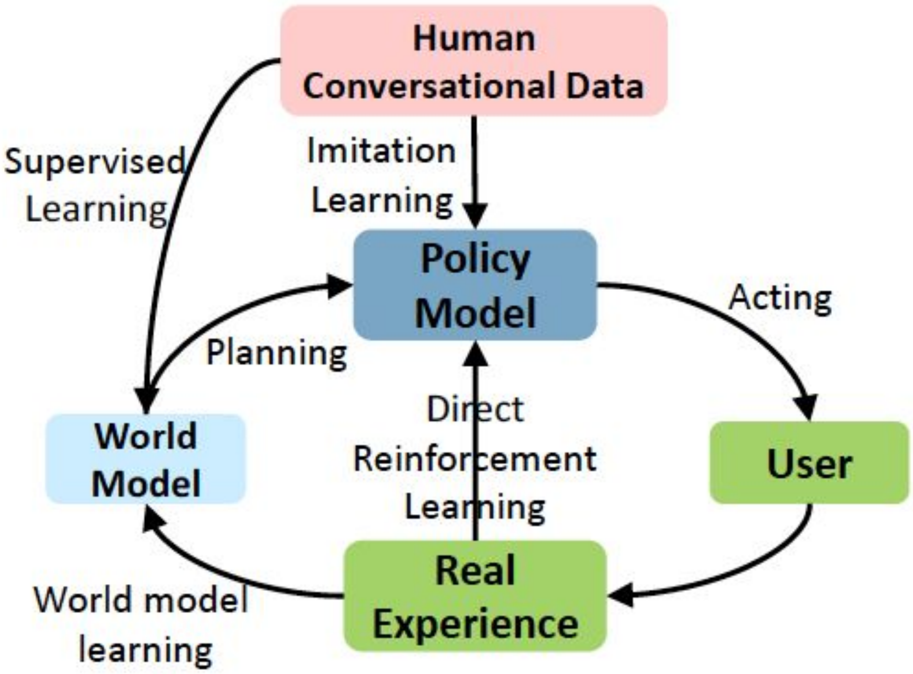
(a) Learning with real users



(b) Learning with user simulators

[15] *Baolin P, et. al., Deep Dyna-Q: Integrating Planning for Task-Completion Dialogue Policy Learning*

2.6 Deep Dyna-Q: Integrating Planning for Task-Completion Dialogue Policy Learning



(c) Learning with real users via DDQ

[15] Baolin P, et. al., Deep Dyna-Q: Integrating Planning for Task-Completion Dialogue Policy Learning

2.6 Deep Dyna-Q: Integrating Planning for Task-Completion Dialogue Policy Learning

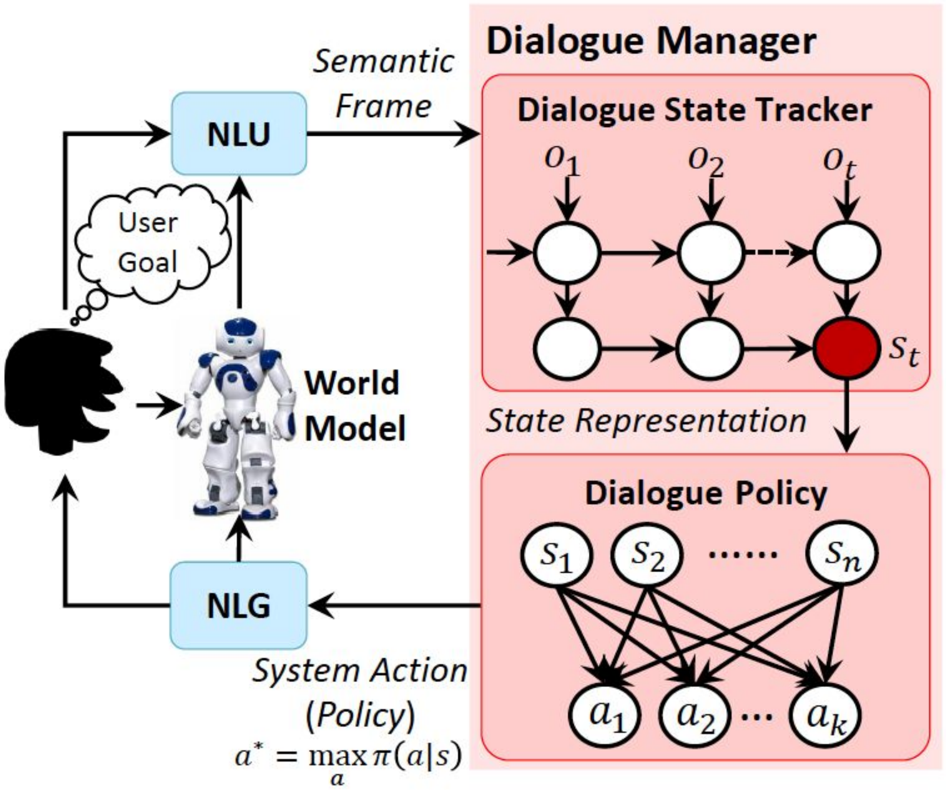


Figure 2: Illustration of the task-completion DDQ dialogue agent.

[15] Baolin P, et. al., Deep Dyna-Q: Integrating Planning for Task-Completion Dialogue Policy Learning

2.6 Deep Dyna-Q: Integrating Planning for Task-Completion Dialogue Policy Learning

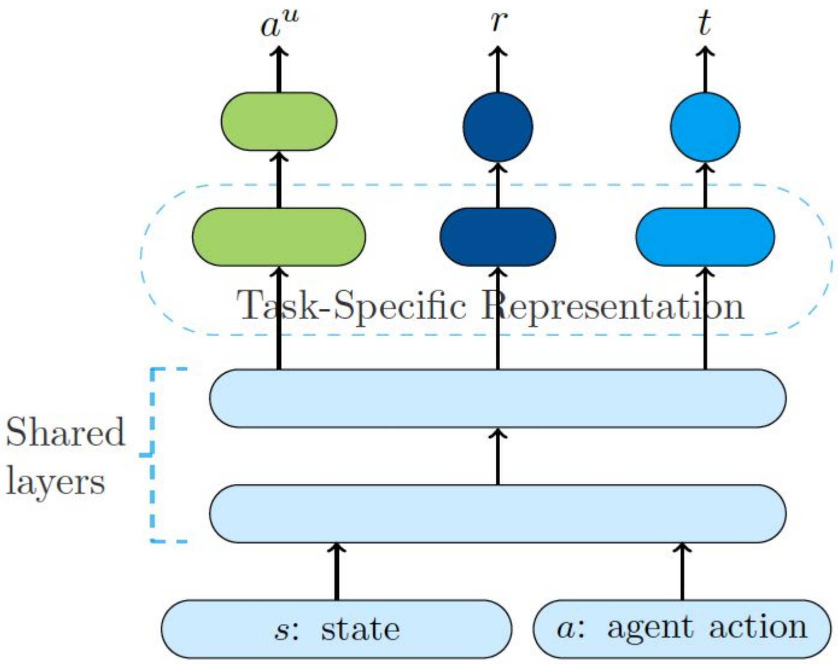


Figure 3: The world model architecture

[15] Baolin P, et. al., Deep Dyna-Q: Integrating Planning for Task-Completion Dialogue Policy Learning

3.
Deep Reinforcement Learning Approaches-III (45min)

Deep Reinforcement Learning Approaches-III:

- Natural Language Models using RL with Emotion Feedback [16]
- Weakly Supervised Dialogue Policy Learning [17]
- Weakly Supervised Method for Topic Segmentation and Labelling in Goal-oriented Dialogues via RL [18]
- Deep Reinforcement Learning for Multi-Domain Dialogue Systems [19]
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3.1 Natural Language Models using RL with Emotion Feedback

Current research in dialogue systems is focused on conversational assistants working on short conversations in either task-oriented or open domain settings. In this paper, we focus on improving task-based conversational assistants online, primarily those working on document-type conversations (e.g., emails) whose contents may or may not be completely related to the assistant's task.

We propose "NARLE" a deep reinforcement learning (RL) framework for improving the natural language understanding (NLU) component of dialogue systems online without the need to collect human labels for customer data.

The proposed solution associates user emotion with the assistant's action and uses that to improve NLU models using policy gradients. For two intent classification problems, we empirically show that using reinforcement learning to fine tune the pre-trained supervised learning models improves performance up to 43%.

Furthermore, we demonstrate the robustness of the method to partial and noisy implicit feedback.

[16] Ruijie Z., et. al., Natural Language Models using RL with Emotion Feedback

3.1 Natural Language Models using RL with Emotion Feedback

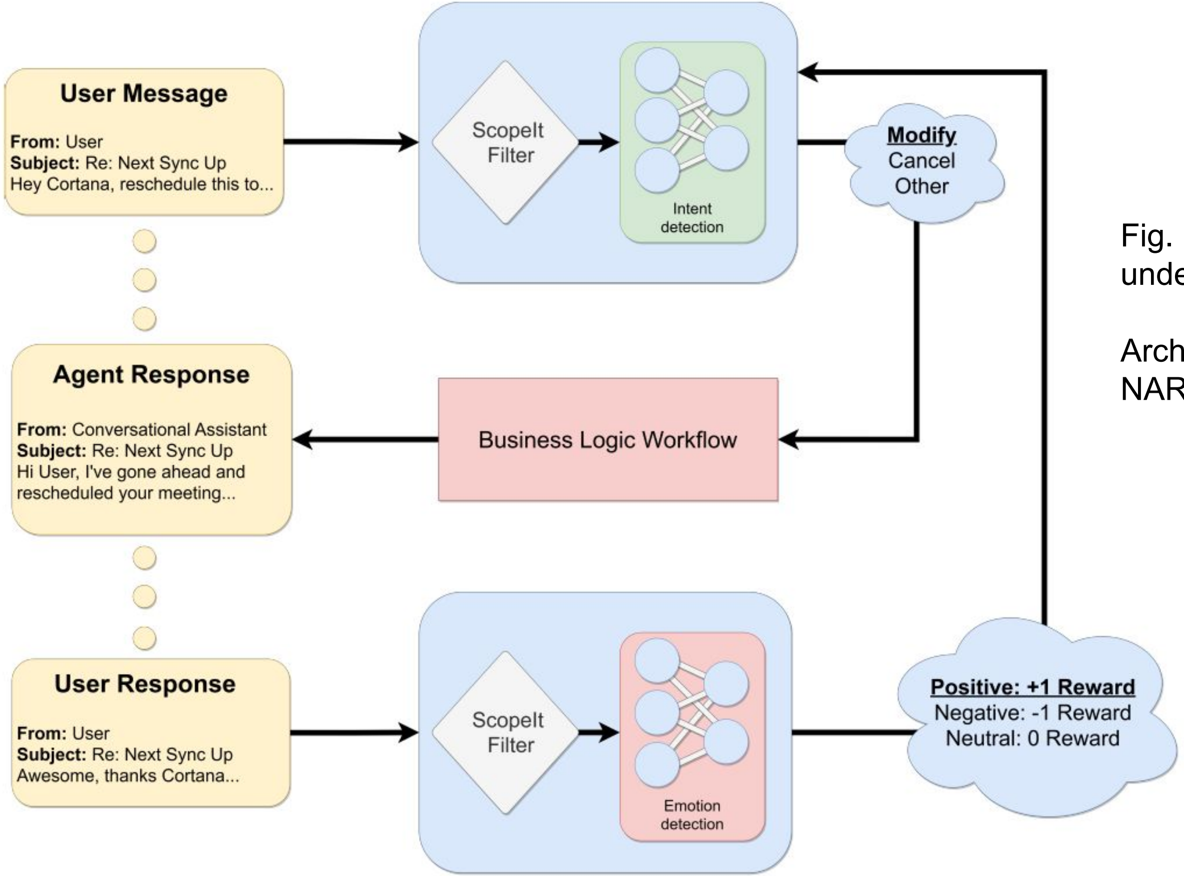

Fig. 1: Improving natural language understanding using implicit feedback

Architecture of proposed framework NARLE

[16] Ruijie Z., et. al., Natural Language Models using RL with Emotion Feedback

3.2 *Weakly Supervised Dialogue Policy Learning: Reward Estimation for Multi-turn Dialogue*

An intelligent dialogue system in a multi-turn setting should not only generate the responses which are of good quality, but it should also generate the responses which can lead to long-term success of the dialogue. Although, the current approaches improved the response quality, but they overlook the training signals present in the dialogue data.

We can leverage these signals to generate the weakly supervised training data for learning dialog policy and reward estimator, and make the policy take actions (generates responses) which can foresee the future direction for a successful (rewarding) conversation.

We simulate the dialogue between an agent and a user (modelled similar to an agent with supervised learning objective) to interact with each other.
The agent uses dynamic blocking to generate ranked diverse responses and exploration exploitation to select among the Top-K responses.

Each simulated state-action pair is evaluated (works as a weak annotation) with three quality modules: Semantic Relevant, Semantic Coherence and Consistent Flow.

Empirical studies with two benchmarks indicate that our model can significantly out-perform the response quality and lead to a successful conversation on both automatic evaluation and human judgment.

[17] Anant K , WeaSuL: Weakly Supervised Dialogue Policy Learning: Reward Estimation for Multi-turn Dialogue

3.2 *Weakly Supervised Dialogue Policy Learning: Reward Estimation for Multi-turn Dialogue*

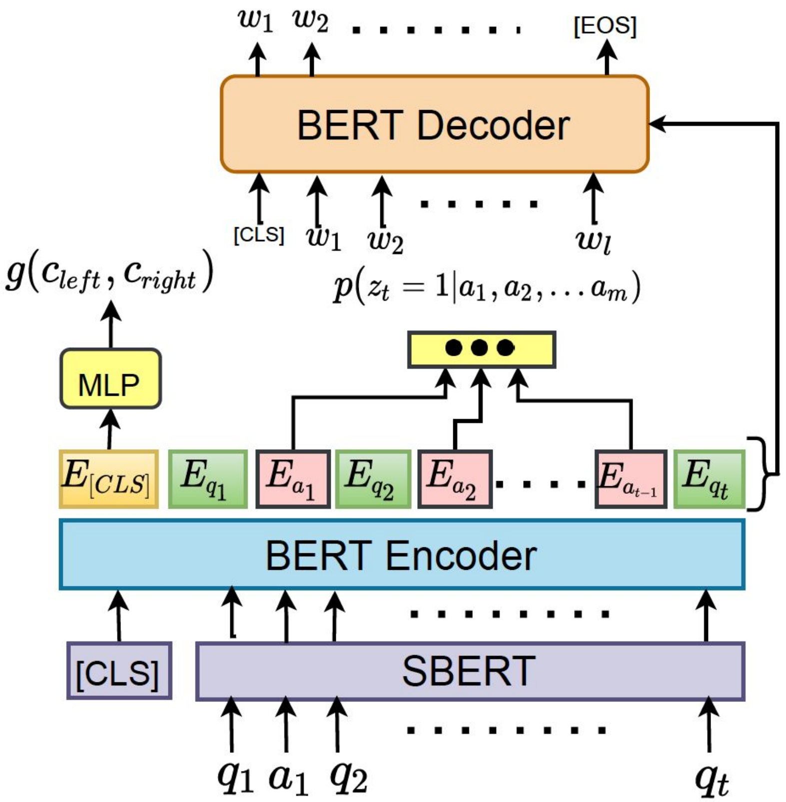

Figure 1: BERT based Encoder-Decoder with Semantic Coherence and Relevance. Similarly, Consistent Flow loss is also calculated using encoder.

[17] Anant K , WeaSuL: Weakly Supervised Dialogue Policy Learning: Reward Estimation for Multi-turn Dialogue

3.3 *A Weakly Supervised Method for Topic Segmentation and Labelling in Goal-oriented Dialogues via RL*

Topic structure analysis plays a pivotal role in dialogue understanding. We propose a reinforcement learning (RL) method for topic segmentation and labeling in goal-oriented dialogues, which aims to detect topic boundaries among dialogue utterances and assign topic labels to the utterances.

We address three common issues in the goal-oriented customer service dialogues: informality, local topic continuity, and global topic structure. We explore the task in a weakly supervised setting and formulate it as a sequential decision problem.

The proposed method consists of a state representation network to address the informality issue, and a policy network with rewards to model local topic continuity and global topic structure. To train the two networks and offer a warm-start to the policy, we firstly use some keywords to annotate the data automatically.

We then pre-train the networks on noisy data. Henceforth, the method continues to refine the data labels using the current policy to learn better state representations on the refined data for obtaining a better policy. Results demonstrate that this weakly supervised method obtains substantial improvements over state-of-the-art baselines.

[18] Ryuichi T, et. al. A Weakly Supervised Method for Topic Segmentation and Labelling in Goal-oriented Dialogues via RL

3.3 *A Weakly Supervised Method for Topic Segmentation and Labelling in Goal-oriented Dialogues via RL*

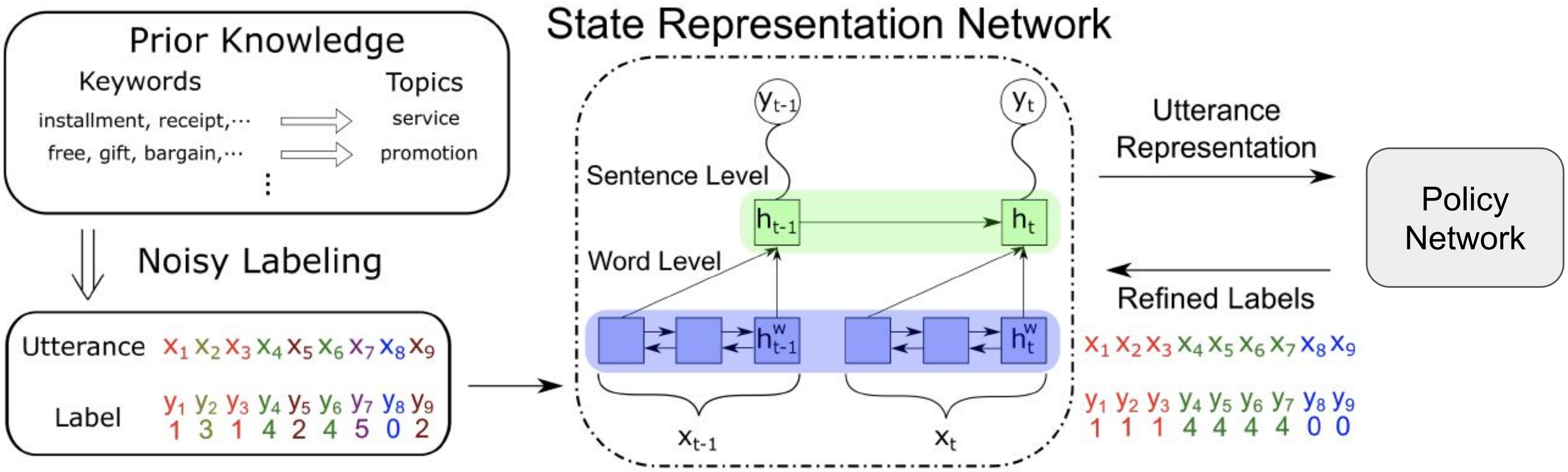

Figure 1: Illustration of the model. SRN adopts a hierarchical LSTM to represent utterances and provides state representations to PN. Data labels are refined to retrain SRN and PN to learn better state representations and policies. The label y and the action a are in the same space.

[18] Ryuichi T, et. al. A Weakly Supervised Method for Topic Segmentation and Labelling in Goal-oriented Dialogues via RL

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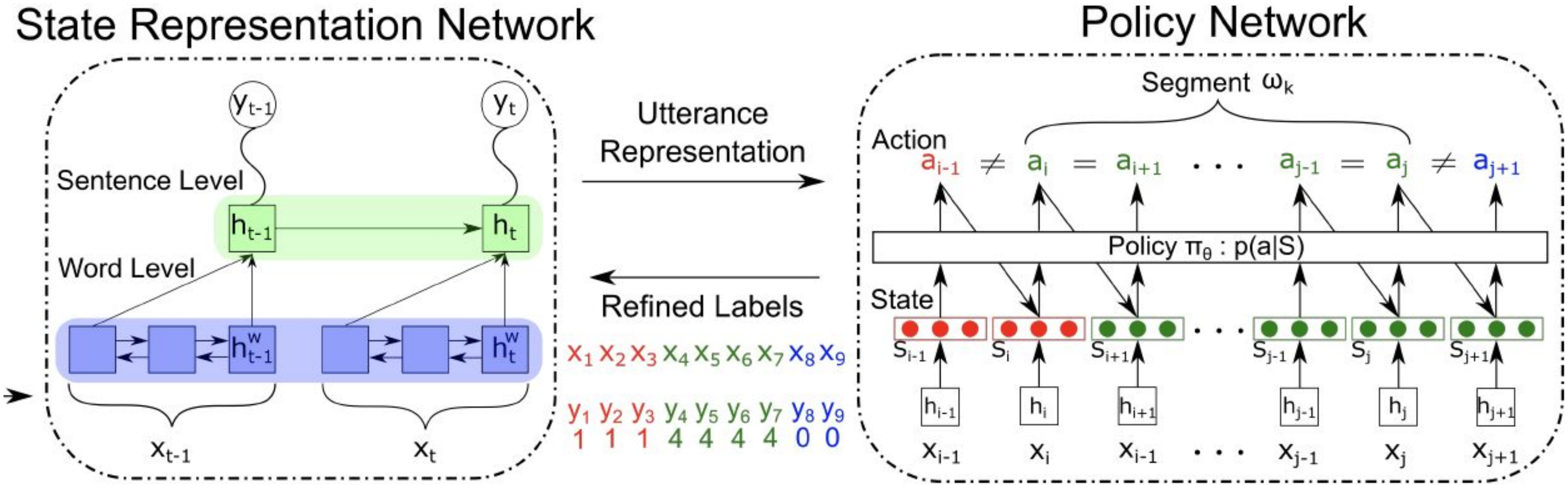

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3.4 Deep Reinforcement Learning for Multi-Domain Dialogue Systems

Standard deep reinforcement learning methods such as Deep Q-Networks (DQN) for multiple tasks (domains) face scalability problems.

We propose a method for multi-domain dialogue policy learning—termed NDQN, and apply it to an information-seeking spoken dialogue system in the domains of restaurants and hotels.

Experimental results comparing DQN (baseline) versus NDQN (proposed) using simulations report that our proposed method exhibits better scalability and is promising for optimising the behaviour of multi-domain dialogue systems.

[19] Heriberto C, et.al. Deep Reinforcement Learning for Multi-Domain Dialogue Systems

3.4 Deep Reinforcement Learning for Multi-Domain Dialogue Systems

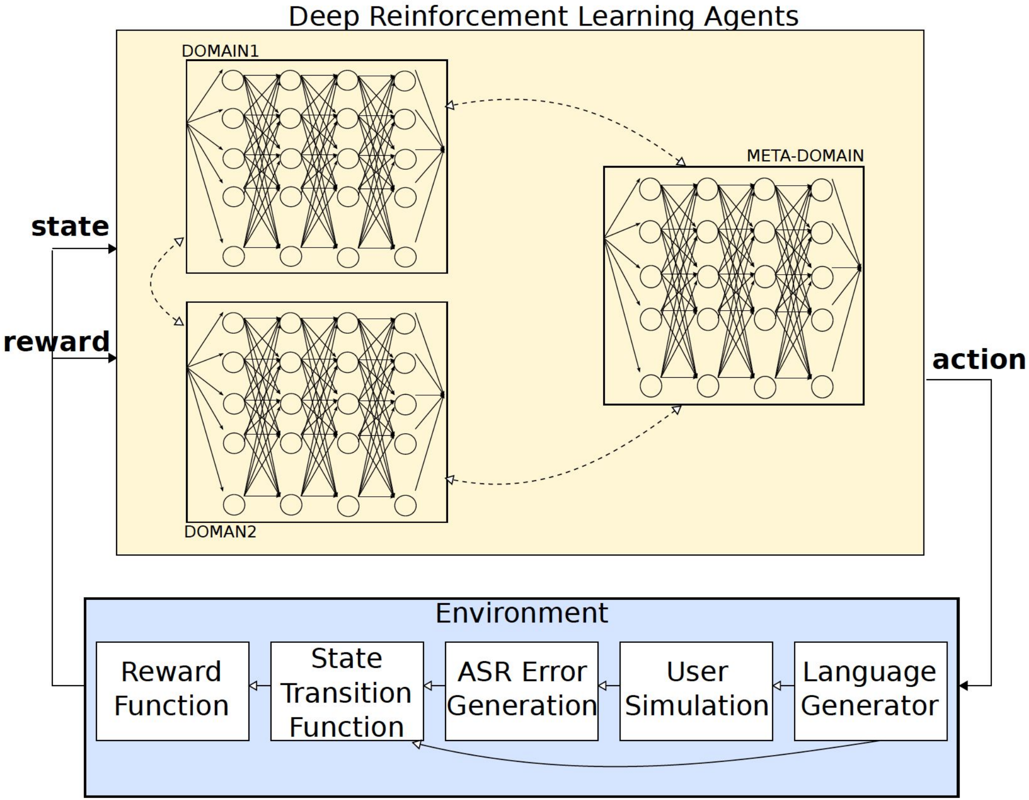

Figure 1: Multi-domain DRL agents with flexible interaction. The dashed arrows connecting domains denote flexible transitions between domains in order to avoid a rigid structure in the interaction.

Although all policies are considered for decision-making, only one domain can be executed at a time implying that a previous domain can continue its execution in order to resume the interaction.

[19] Heriberto C, et.al. Deep Reinforcement Learning for Multi-Domain Dialogue Systems

3.5 Decision Transformer: Reinforcement Learning via Sequence Modeling

We introduce a framework that abstracts Reinforcement Learning (RL) as a sequence modeling problem. This allows us to draw upon the simplicity and scalability of the Transformer architecture, and associated advances in language modeling such as GPT-x and BERT.

In particular, we present Decision Transformer, an architecture that casts the problem of RL as conditional sequence modeling. Unlike prior approaches to RL that fit value functions or compute policy gradients, Decision Transformer simply outputs the optimal actions by leveraging a causally masked Transformer.

By conditioning an autoregressive model on the desired return (reward), past states, and actions, our Decision Transformer model can generate future actions that achieve the desired return.

Despite its simplicity, Decision Transformer matches or exceeds the performance of state-of-the-art model-free offline RL baselines on Atari, OpenAI Gym, and Key-to-Door tasks.

[20] Lili C., et. al., Decision Transformer: Reinforcement Learning via Sequence Modeling

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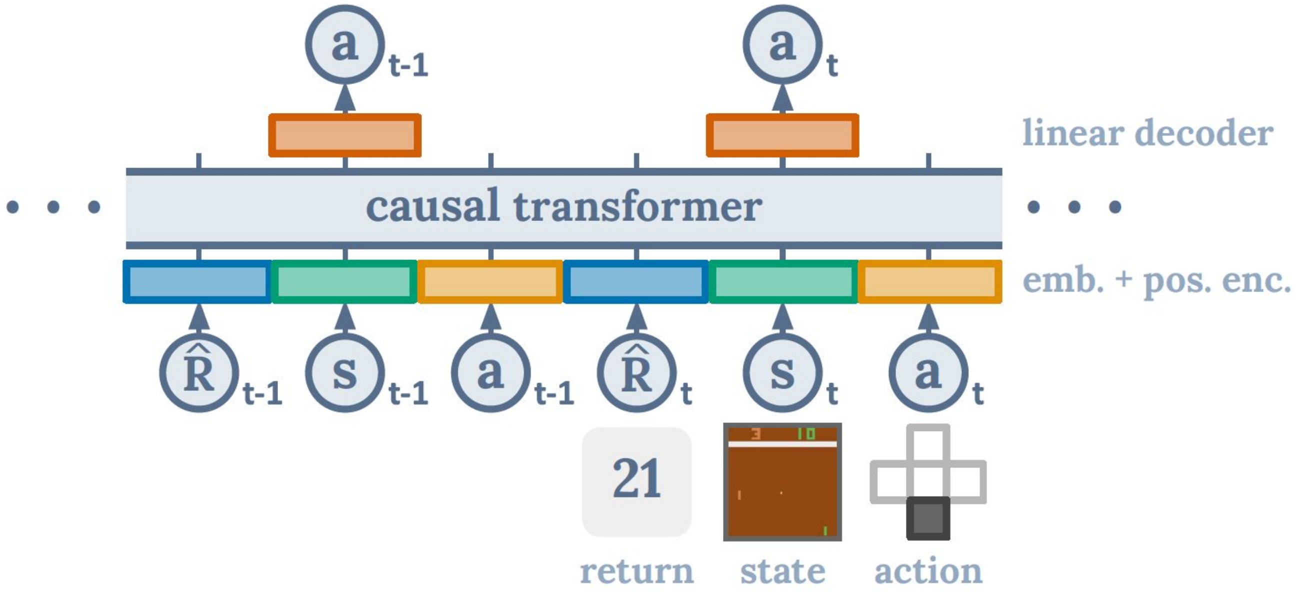

Figure 1: Decision Transformer architecture. States, actions, and returns are fed into modality specific linear embeddings and a positional episodic timestep encoding is added. Tokens are fed into a GPT architecture which predicts actions autoregressively using a causal self-attention mask.

[20] Lili C., et. al., Decision Transformer: Reinforcement Learning via Sequence Modeling

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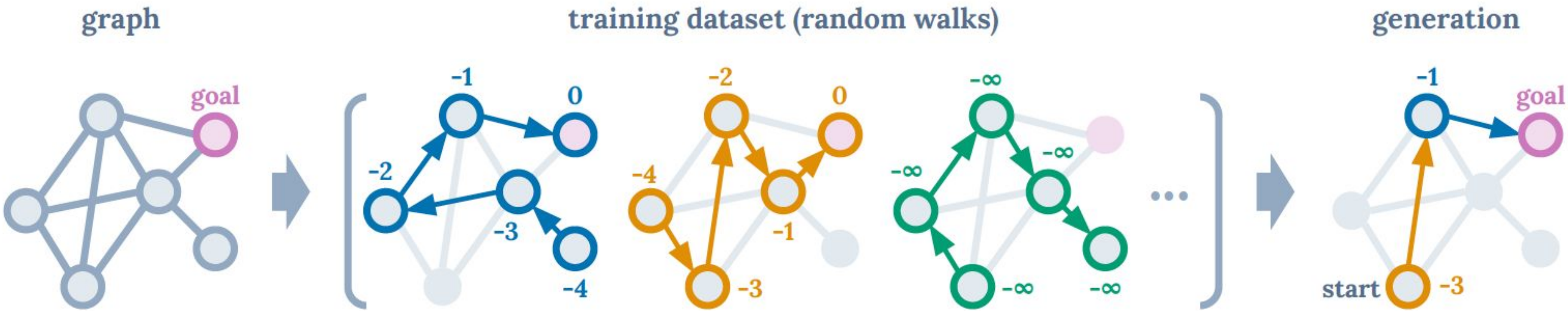

Figure 2: Illustrative example of finding shortest path for a fixed graph (left) posed as reinforcement learning. Training dataset consists of random walk trajectories and their per-node returns-to-go (middle). Conditioned on a starting state and generating largest possible return at each node, Decision Transformer sequences optimal paths.

[20] Lili C., et. al., Decision Transformer: Reinforcement Learning via Sequence Modeling

Thank You.

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